

Mekan Myradov

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EDUCATION

- **Sabanci University** Istanbul, Turkey
MSc in Data Science; GPA: 3.54 *Sep 13, 2022 - Jun 2026 (Expected)*
Courses: Machine Learning, Pattern Recognition, Deep Learning, Network Science, Big Data Processing
Statistical Modelling, Statistical Inference, Engineering Optimization.
- **Kirklareli University** Kirklareli, Turkey
BSc in Software Engineering; GPA: 3.71 *Sep 10, 2018 - Aug 3, 2022*

PUBLICATIONS

- **A Systematic Evaluation of Single-Cell Batch Integration Metrics and sBEE: A Robust New Metric**
Mekan Myradov, Aissa Houdjedj, Oznur Tastan, and Hilal Kazan
Under review for European Conference on Computational Biology (ECCB), 2026.
- **SCITUNA: Single-Cell data integration tool using network alignment**, BMC Bioinformatics, 2025.
Aissa Houdjedj, Yacine Marouf, Mekan Myradov, Onur Dogan, Burak Onur Erten, Oznur Tastan, Cesim Erten and Hilal Kazan
[PDF] / [Code]

ACADEMIC PROJECTS

- **Single-Cell Multiome Analysis (Dec '25)**
A comprehensive deep learning pipeline for analyzing single-cell multiome data, benchmarking multiple state-of-the-art integration methods including CCA, Joint VAE, MultiVI, MOFA+ and Seurat (Weighted Nearest Neighbors).
[PDF] / [Code]
- **A Toolbox for ANOVA and Linear Regression (May '25)**
Comprehensive statistical analysis toolkit that implements
 - One-Way ANOVA (group comparison, contrast analysis)
 - Two-Way ANOVA (two-factor analysis with interaction effects)
 - Multiple Linear Regression (parameter estimation, hypothesis testing, prediction intervals)
 - Multiple comparison corrections (Bonferroni, Sidak, Scheffe, Tukey) with individual and simultaneous confidence intervals[PDF] / [Code]
- **Synthetic Data Generation with Gaussian Mixture Models (Sep '23)**
Implemented a synthetic data generation algorithm in Python using Gaussian Mixture Models
[PDF] / [Code]

ACADEMIC ACTIVITIES

- **Int. Symposium on Health Informatics and Bioinformatics (HIBIT)** Istanbul, Turkey
Talk *Oct 31, 2025 - Nov 1, 2025*
- **Int. Conf. on Research in Computational Molecular Biology (RECOMB)** Istanbul, Turkey
Participant *Apr 16, 2023 - Apr 19, 2023*
- **Workshop on the analysis and integration of single-cell RNA-seq datasets** Antalya, Turkey
Participant *Mar 23, 2023*

EXPERIENCE

- **Sabanci University** Istanbul, Turkey
Teaching Assistant *Sep 2022 - Current*
 - **Courses:** Software Engineering, Advanced Programming
 - Held weekly recitation session and office hours.
 - Graded programming homework and examinations.
 - Helped students so they can follow software development lifecycle best practices.
- **Kirklareli University** Kirklareli, Turkey
Research Intern - Asst. Prof. Edip Serdar GÜNER *Nov 2021 - Jun 2022*
 - Participated in the project that was about the creation and analysis of a Turkish drug reviews dataset from social media.
 - Created a dataset by compiling Turkish drug reviews from social media.

- Analyzed dataset using text mining methods (Zemberek-NLP and VNLP).
- Sentiment Analysis with Hugging Face models (Turkish BERT etc).

• **Cognitiwe AI**

Research Intern

- Dealt with feature detectors for images
- Worked with SIFT (Scale Invariant Feature Transform) method
- Worked with SURF (Speeded-Up Robust Features) method
- Worked with ORB (Oriented FAST and Rotated BRIEF) method

Istanbul, Turkey

May 2021 - Sep 2021

• **Mimix AI**

Research Intern

- Worked with face swap methods especially DeepFaceLab components
- Worked with CNN-based face detection methods (S3FD)
- Worked with CNN-based face alignment methods (2D-FAN)
- Worked with CNN-based face segmentation methods (TernausNet)
- Used autoencoders to learn mappings between faces

Istanbul, Turkey

Dec 2020 - Apr 2021

CERTIFICATIONS AND COURSES

- Machine Learning Specialization (Coursera – Andrew Ng).
- Deep Learning Specialization (Coursera – Andrew Ng).

HONORS AND AWARDS

- TUBITAK Research Scholarship, 2022-2025.
- Sabanci University Tuition Waiver Scholarship, 2022-2026.
- Kirlklareli University – High Honor Graduate
- Success Scholarship for outstanding academic performance, 2019-2022.
- National Olympiad in Informatics 2017 – awardee
- National Olympiad in Informatics 2016 – awardee

SKILLS SUMMARY

- **Software Development Lifecycle:** Agile Methodologies (Scrum), Coding Standards, Code Reviews, Resource Management, Development Processes, Deployment, Testing and Operations
- **Programming Languages:** Python, R, C, C++, Java, SQL, Unix/Shell Scripting
- **Libraries:** NLTK, OpenCV, NumPy, SciPy, Pandas, Matplotlib, Seaborn
- **Machine Learning & Deep Learning:** Spark MLlib, Scikit-learn, TensorFlow, PyTorch, CNNs, RNNs, Transformers, Autoencoders, Gaussian Mixture Models
- **Big Data & Cloud Technologies:** AWS, Hadoop, Apache Spark, Snowflake
- **Development Tools & Platforms:** GIT, Jira, Linux, Unix
- **Text Processing & NLP Libraries:** Zemberek, VNLP, Hugging Face Transformers
- **Domain Expertise:**
 - Computer Vision (SIFT, SURF, ORB, Face Detection/Alignment/Segmentation)
 - Natural Language Processing (BERT, Sentiment Analysis, Text Mining)
 - Computational Biology & Bioinformatics (Single-cell RNA-seq analysis)

SERVICE

- **Military Service:** Feb 2024 - Feb 2025
- **Reviewing:** Bioinformatics